



Data and Artificial Intelligence

Cyber Shujaa Program

Week 3 Assignment

Titanic Exploratory Data Analysis in Python
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1.0 Introduction

The RMS Titanic disaster remains one of the most infamous maritime tragedies in history. On April 15, 1912, the "unsinkable" ship collided with an iceberg during its maiden voyage, resulting in the loss of 1,502 lives out of 2,224 passengers and crew. This exploratory data analysis (EDA) investigates the factors that influenced passenger survival rates using the famous Titanic dataset from Kaggle.

Objectives

The primary objectives of this analysis are to:

- Understand the dataset structure and identify data quality issues
- Explore relationships between features and survival outcomes
- Handle missing values and outliers appropriately
- Identify key factors that influenced survival rates
- Prepare the dataset for future predictive modeling

This analysis follows a structured approach using object-oriented programming principles in Python, employing libraries such as pandas, numpy, matplotlib, seaborn, and scipy for comprehensive data exploration and visualization.

2.0 Task Completion

2.1 Initial Data Exploration

The initial exploration phase involved understanding the dataset's structure, dimensions, and basic characteristics.

INITIAL DATA EXPLORATION													
Dataset Shape: 891 rows x 12 columns													
First 5 rows of the dataset:													
	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S	
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S	

```
Data Types Summary:
int64      5
object     5
float64    2
Name: count, dtype: int64
```

Detailed Column Information:

	Column	Data Type	Non-Null Count	Null Count	Null %	Unique Values
PassengerId	PassengerId	int64	891	0	0.00	891
Survived	Survived	int64	891	0	0.00	2
Pclass	Pclass	int64	891	0	0.00	3
Name	Name	object	891	0	0.00	891
Sex	Sex	object	891	0	0.00	2
Age	Age	float64	714	177	19.87	88
SibSp	SibSp	int64	891	0	0.00	7
Parch	Parch	int64	891	0	0.00	7
Ticket	Ticket	object	891	0	0.00	681
Fare	Fare	float64	891	0	0.00	248
Cabin	Cabin	object	204	687	77.10	147
Embarked	Embarked	object	889	2	0.22	3

The dataset contains various feature types:

- i. **Numerical:** Age, SibSp, Parch, Fare
- ii. **Categorical:** Survived, Pclass, Sex, Embarked
- iii. **Text:** Name, Ticket
- iv. **Mixed:** Cabin (contains both deck letter and room number)

Statistical Summary (Numerical Features):

	count	mean	std	min	25%	50%	75%	max
PassengerId	891.0	446.000000	257.353842	1.00	223.5000	446.0000	668.5	891.0000
Survived	891.0	0.383838	0.486592	0.00	0.0000	0.0000	1.0	1.0000
Pclass	891.0	2.308642	0.836071	1.00	2.0000	3.0000	3.0	3.0000
Age	714.0	29.699118	14.526497	0.42	20.1250	28.0000	38.0	80.0000
SibSp	891.0	0.523008	1.102743	0.00	0.0000	0.0000	1.0	8.0000
Parch	891.0	0.381594	0.806057	0.00	0.0000	0.0000	0.0	6.0000
Fare	891.0	32.204208	49.693429	0.00	7.9104	14.4542	31.0	512.3292

Categorical Features Summary:

Sex – Unique values: 2

Sex

male 577

female 314

Name: count, dtype: int64

Embarked – Unique values: 3

Embarked

S 644

C 168

Q 77

Name: count, dtype: int64

2.2 Data Quality Assessment

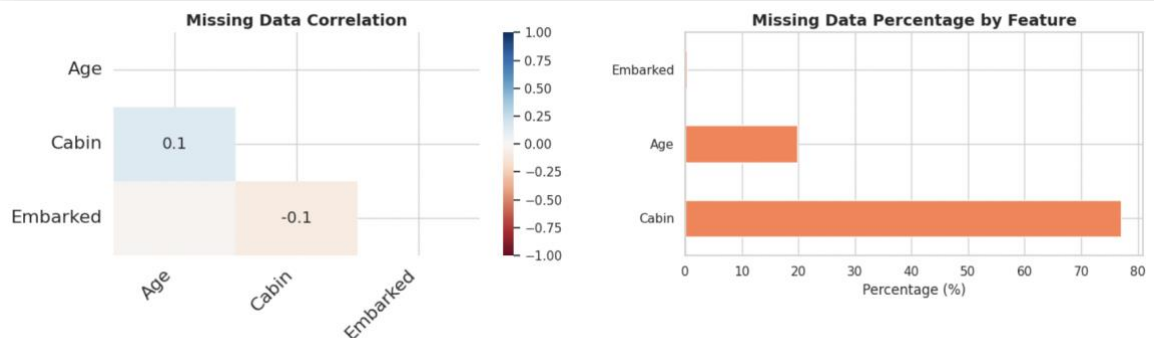
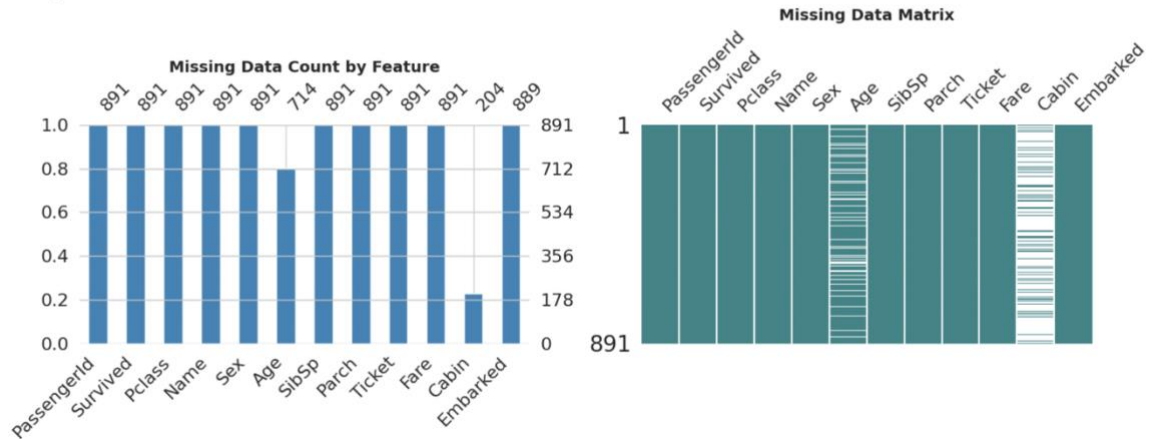
A comprehensive data quality check was performed to identify potential issues.

DATA QUALITY ASSESSMENT

Duplicate rows found: 0

No major data consistency issues found!

Missing Data Visualization:



Key Findings:

- i. No duplicate records were found in the dataset
- ii. Missing values were present in three columns:
 - o Cabin: 77.1% missing (687 out of 891)
 - o Age: 19.9% missing (177 out of 891)
 - o Embarked: 0.2% missing (2 out of 891)
- iii. No major data consistency issues were identified

2.3 Handling Missing Values

Different strategies were employed for handling missing values based on the nature and extent of missingness.

```
=====
MISSING VALUES HANDLING
=====
```

Missing Values Analysis:

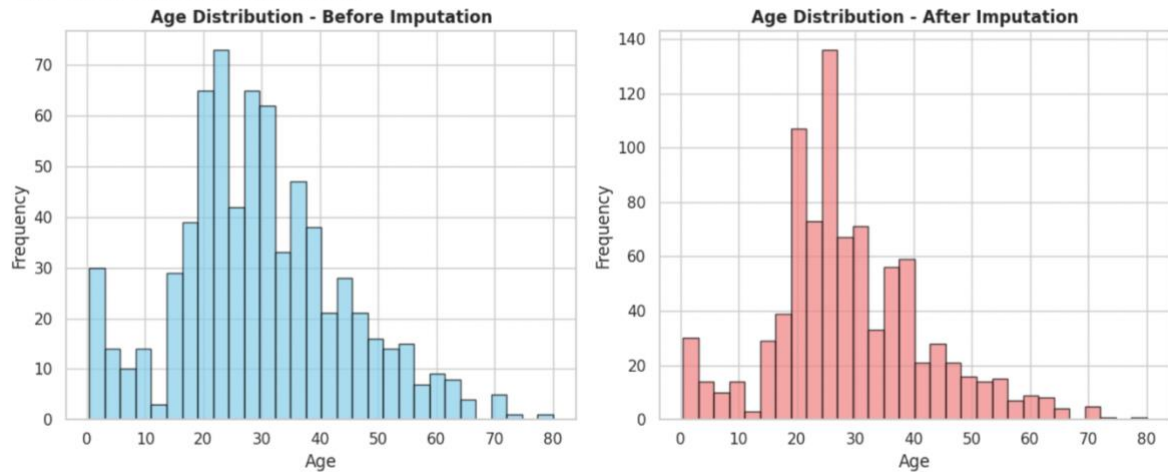
	Column	Missing_Count	Missing_Percentage	Data_Type
Cabin	Cabin	687	77.10	object
Age	Age	177	19.87	float64
Embarked	Embarked	2	0.22	object

Missing Value Strategies:

1. **Age:** Imputed using grouped median by Pclass and Sex
 - Justification: Age likely varies by passenger class and gender
 - Method: Group-wise median imputation for more accurate estimates
2. **Embarked:** Imputed with mode (most frequent value)
 - Justification: Only 2 missing values; mode imputation is appropriate
 - Result: Filled with 'S' (Southampton)
3. **Cabin:** Created indicator variable and extracted deck information
 - Justification: High missingness (77.1%) makes imputation unreliable
 - Created features: 'Has_Cabin' (binary) and 'Deck' (categorical)

Missing Value Handling Report:
 Age: Imputed with median by Pclass and Sex
 Embarked: Imputed with mode: S
 Cabin: Created Has_Cabin indicator and Deck feature

Imputation Impact Visualization:

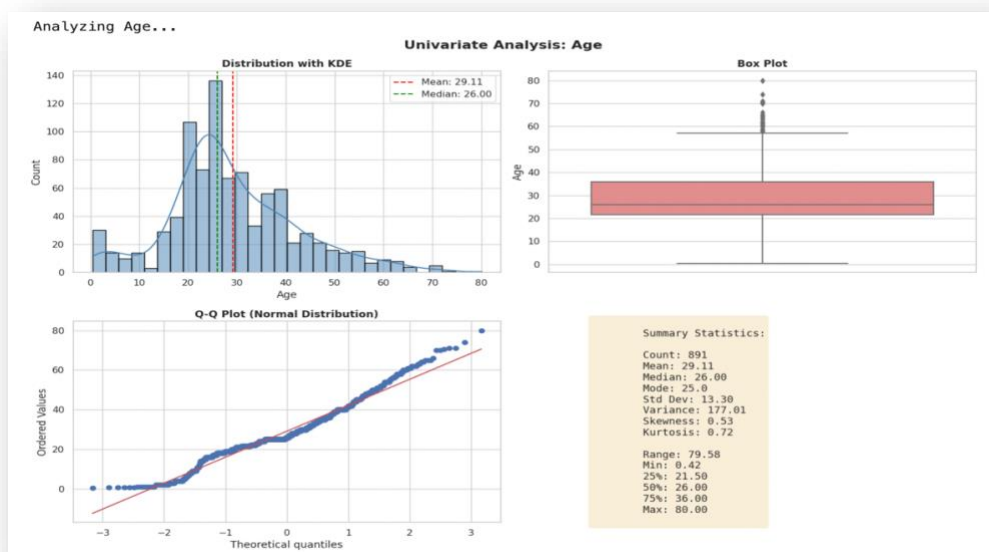


Missing values after handling:
 Cabin 687
 dtype: int64

2.4 Univariate Analysis

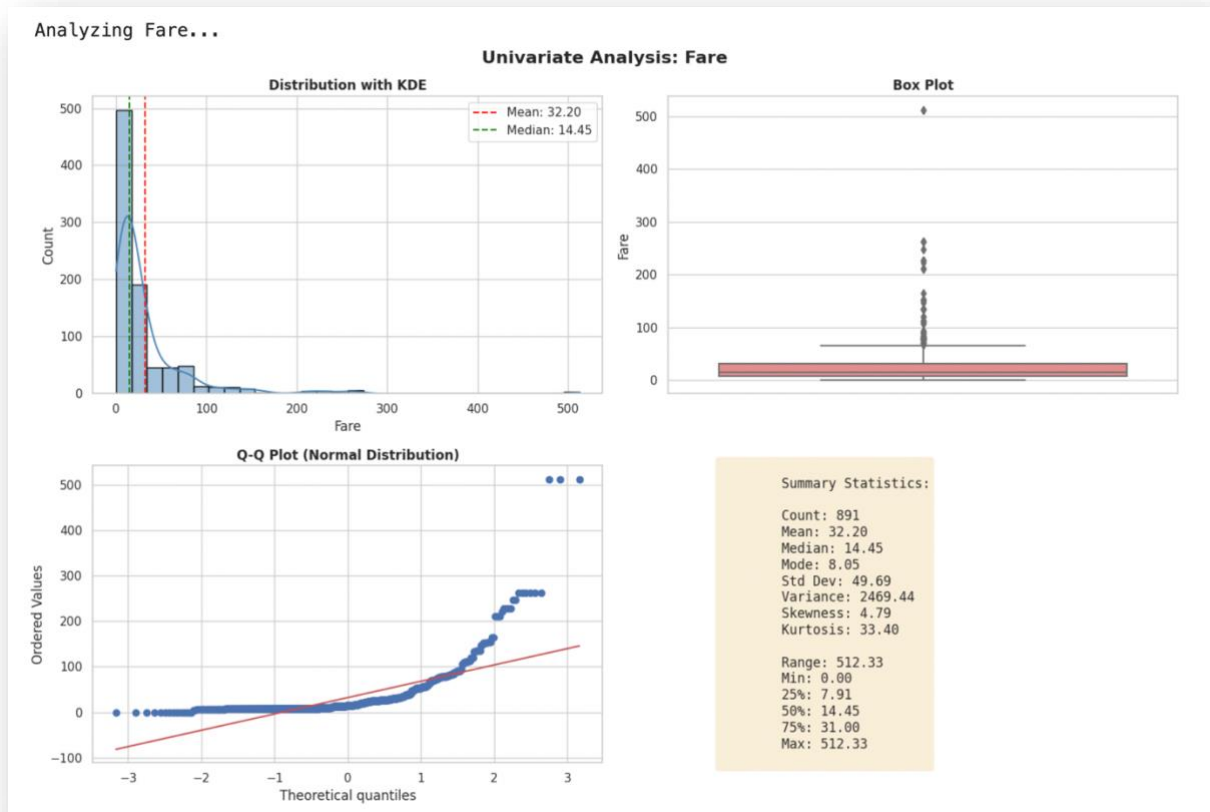
Each feature was analyzed individually to understand its distribution and characteristics.

Numerical Features Analysis



Age Distribution:

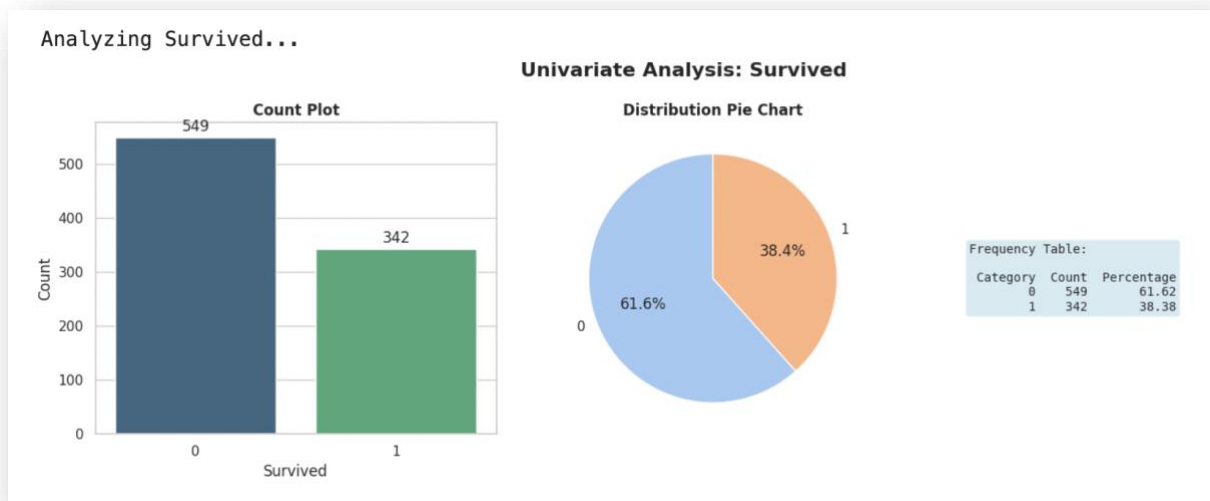
- i. Mean: 29.7 years
- ii. Median: 28.0 years
- iii. Right-skewed distribution with most passengers between 20-40 years
- iv. Presence of infant passengers and elderly (up to 80 years)



Fare Distribution:

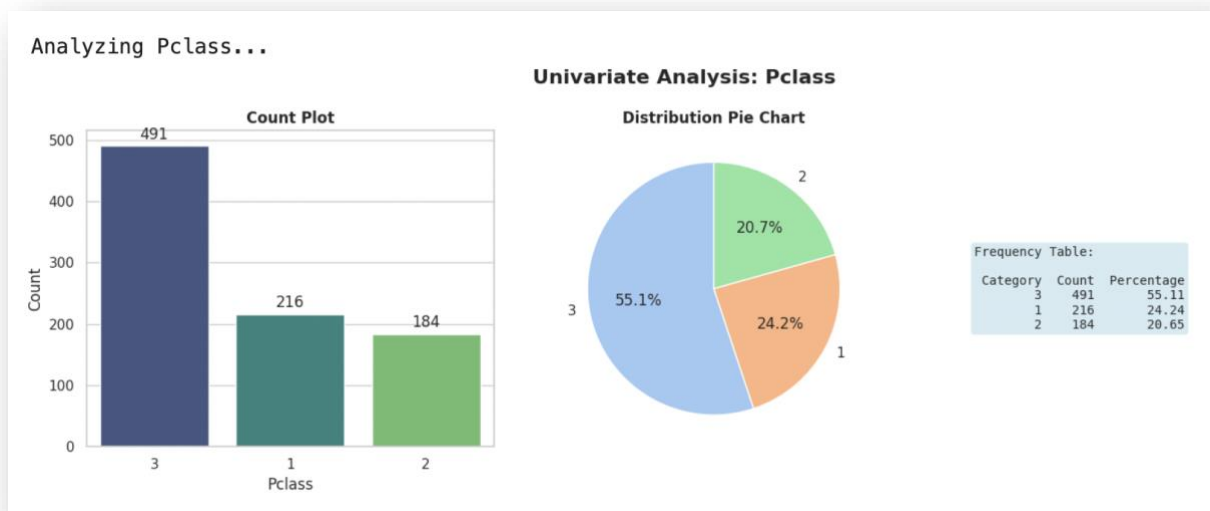
- i. Highly right-skewed with extreme outliers
- ii. Median fare: £14.45
- iii. Maximum fare: £512.33 (luxury accommodations)
- iv. Many passengers paid minimal fares (3rd class)

Categorical Features Analysis



Survived Distribution:

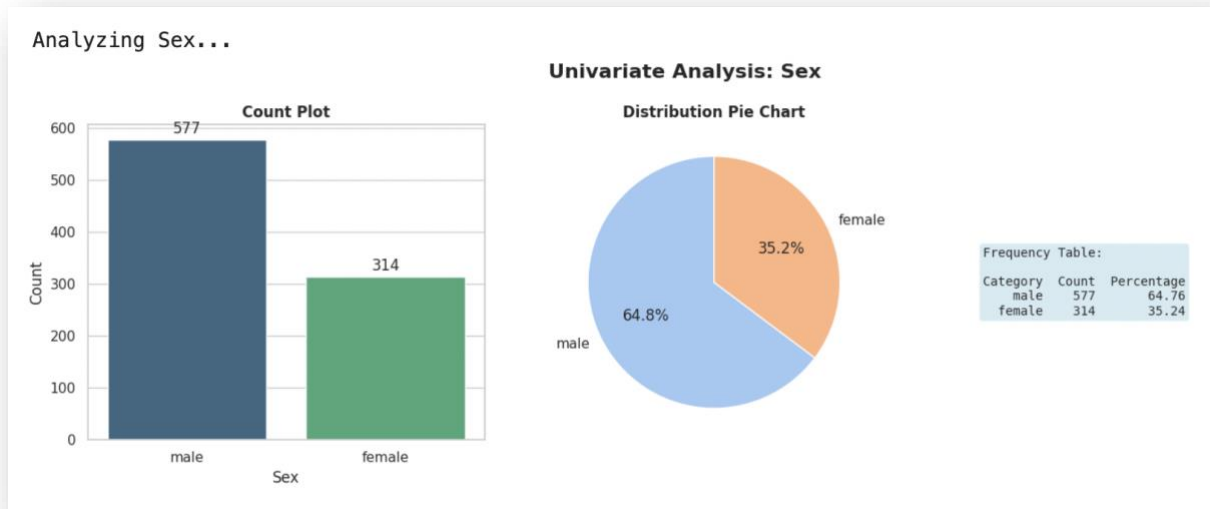
- i. Did not survive (0): 549 (61.6%)
- ii. Survived (1): 342 (38.4%)



Passenger Class Distribution:

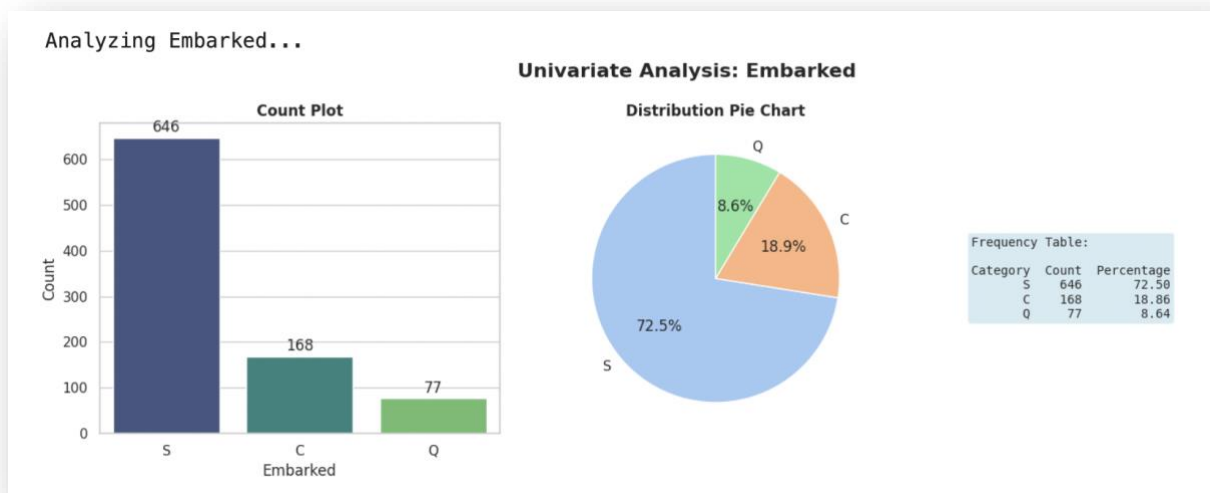
- i. 3rd Class: 491 passengers (55.1%)
- ii. 1st Class: 216 passengers (24.2%)

iii. 2nd Class: 184 passengers (20.7%)



Sex Distribution:

- Male passengers: 577 (64.8%)
- Female passengers: 314 (35.2%)



Embarked Distribution:

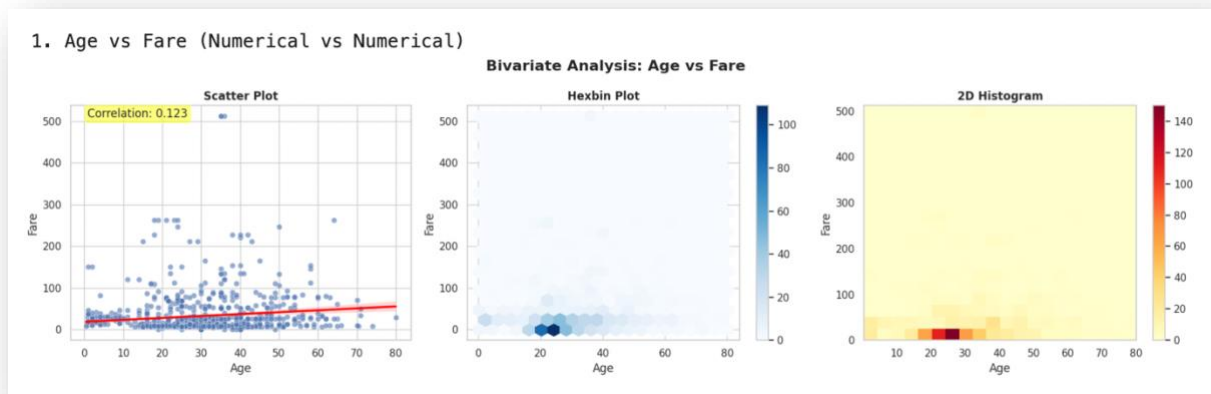
- Category S: 646 (72.5%)
- Category C: 168 (18.9%)

iii. Category Q: 77 (8.6%)

2.5 Bivariate Analysis

Relationships between pairs of features were explored to identify patterns and correlations.

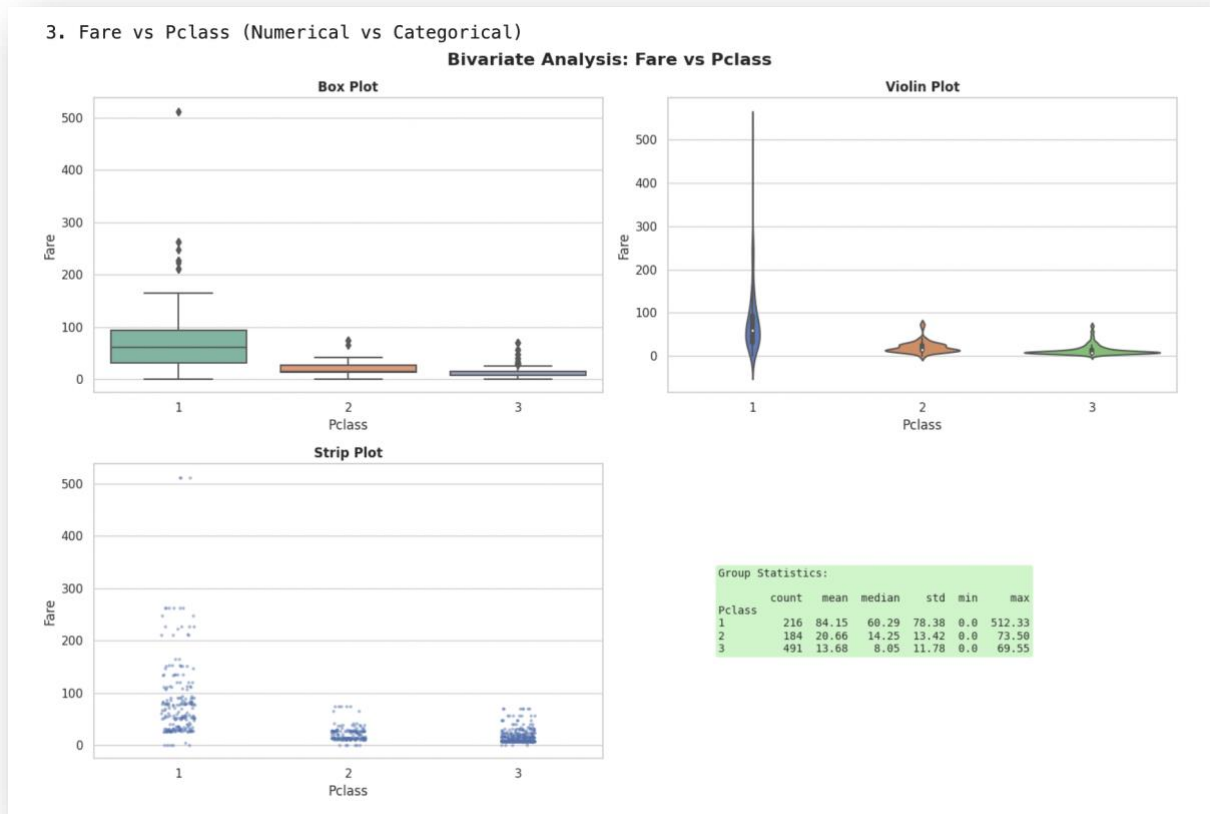
Numerical vs Numerical Analysis



Key Findings:

- i. Weak positive correlation between Age and Fare ($r = 0.096$)
- ii. Higher fares generally associated with older passengers

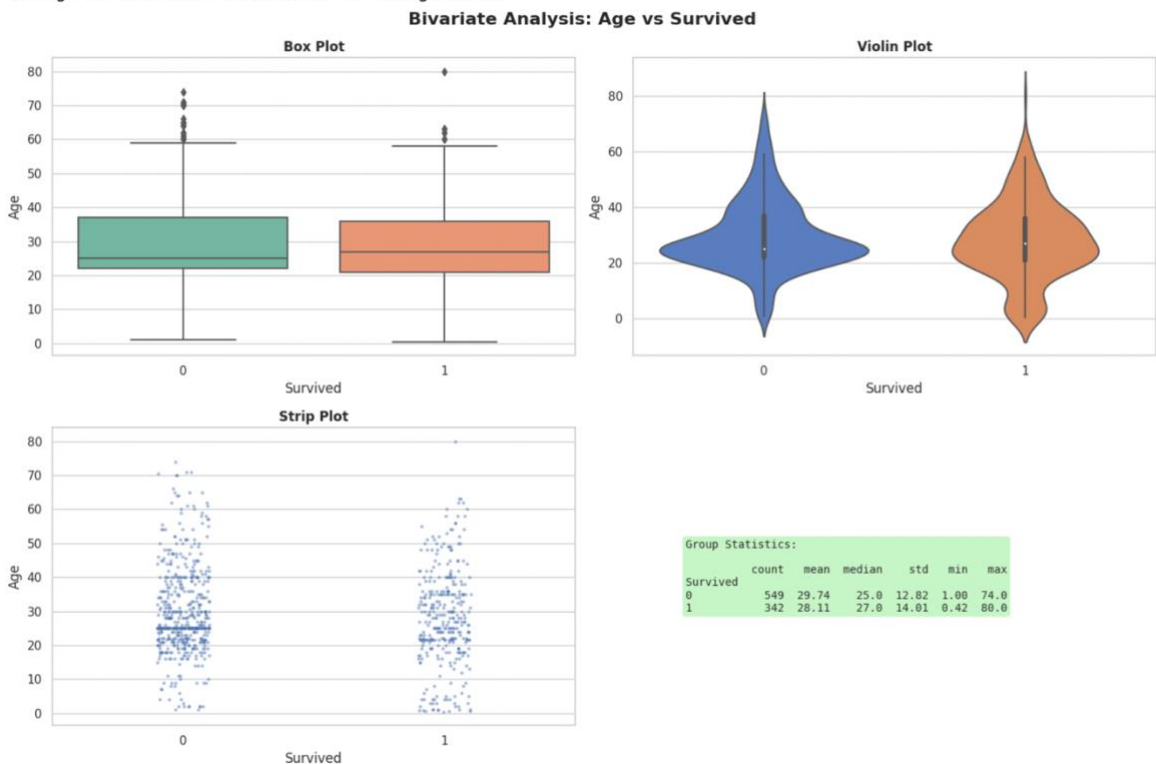
Numerical vs Categorical Analysis



Fare by Passenger Class:

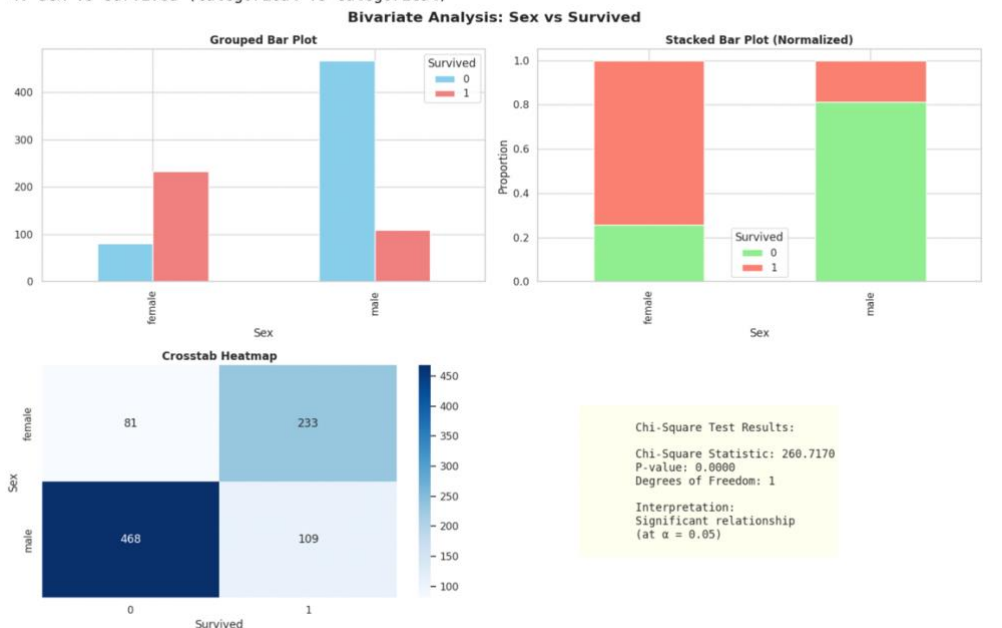
- Clear separation in fare ranges by class
- 1st class: Median ~£60, with high variability
- 2nd class: Median ~£15
- 3rd class: Median ~£8, with minimal variation

2. Age vs Survived (Numerical vs Categorical)



Categorical vs Categorical Analysis

4. Sex vs Survived (Categorical vs Categorical)

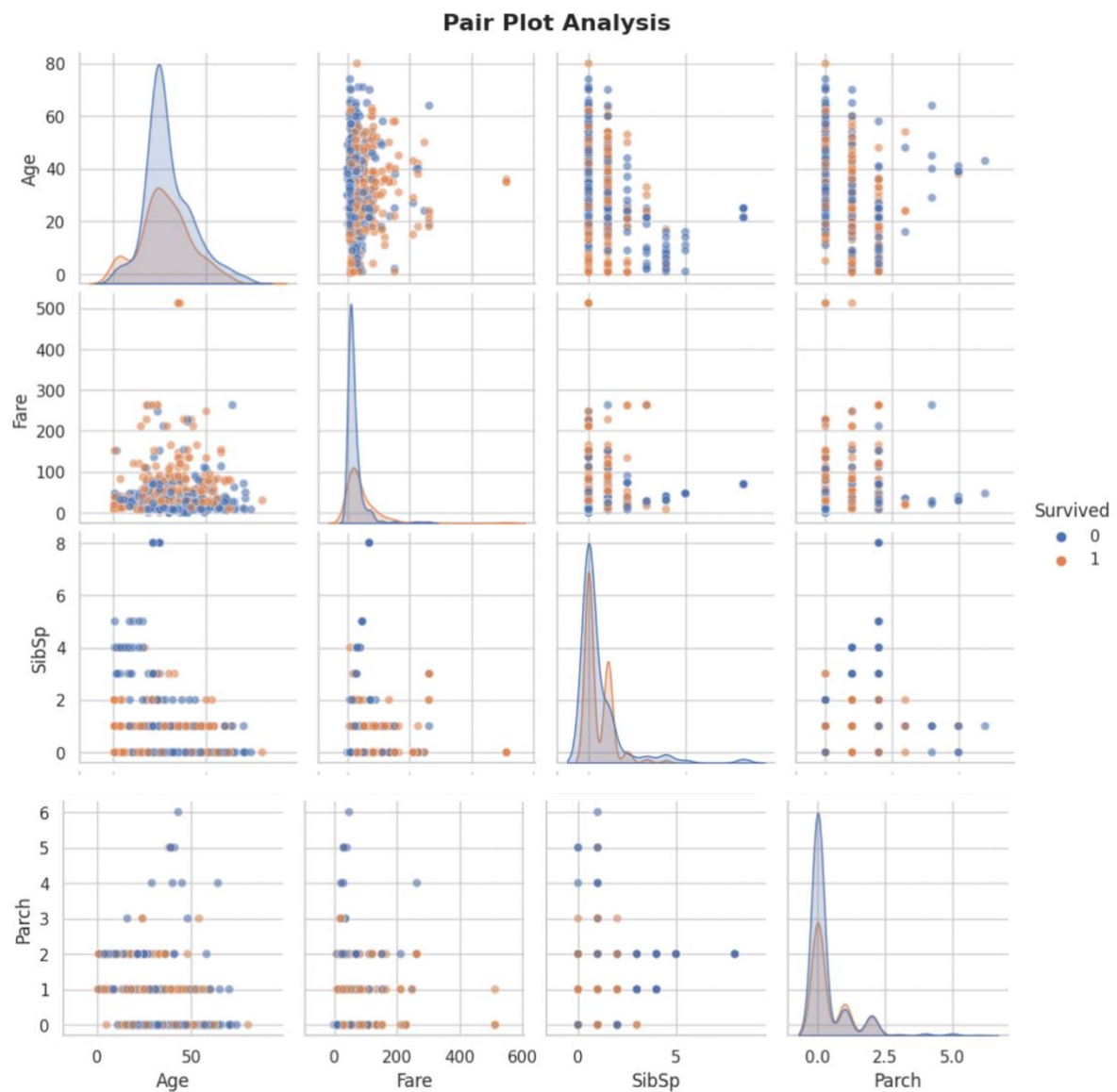


Survival by Gender:

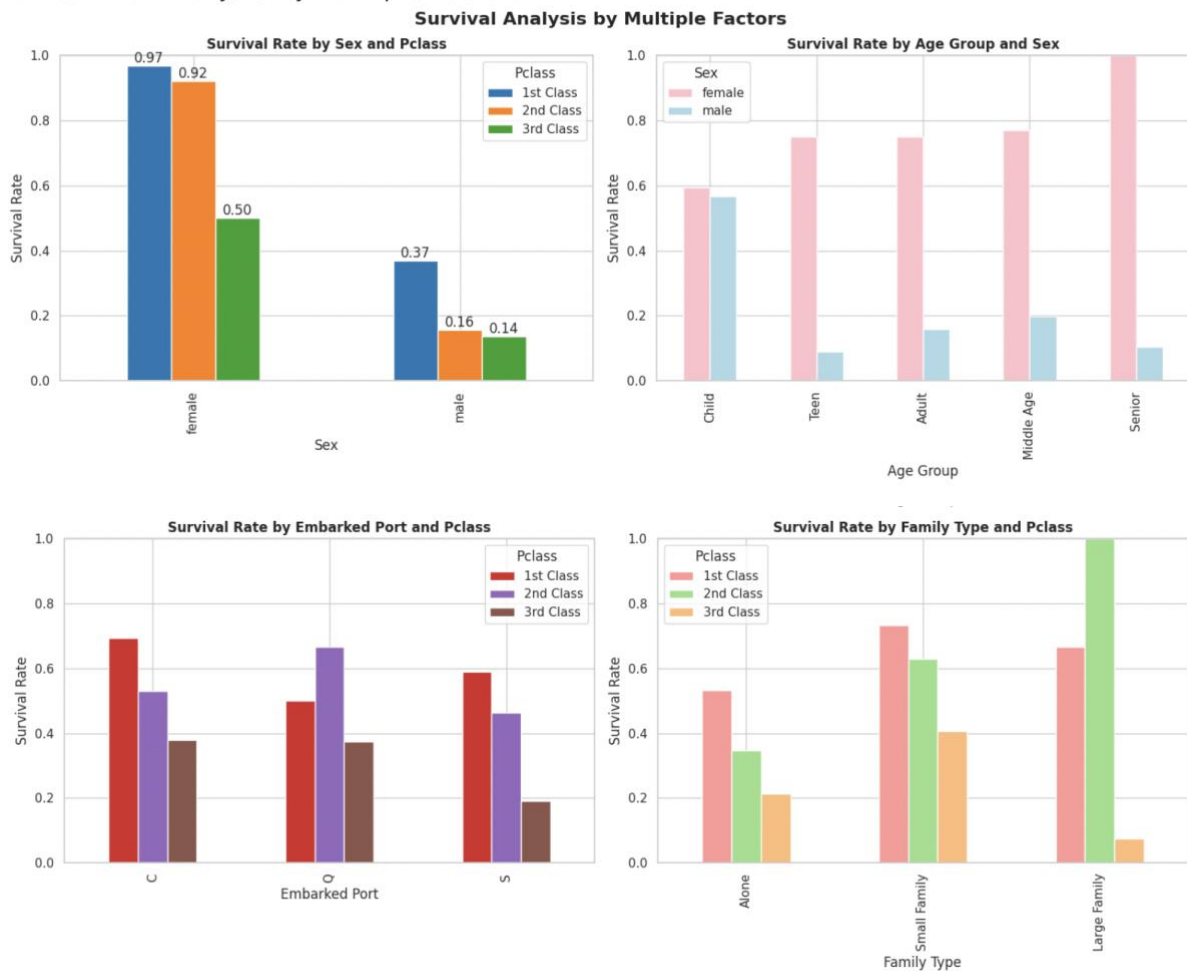
- i. Female survival rate: 74.2%
- ii. Male survival rate: 18.9%
- iii. Strong evidence of "women and children first" protocol

2.6 Multivariate Analysis

Complex relationships between multiple variables were explored simultaneously.



3. Survival Analysis by Multiple Factors



Key Multivariate Findings:

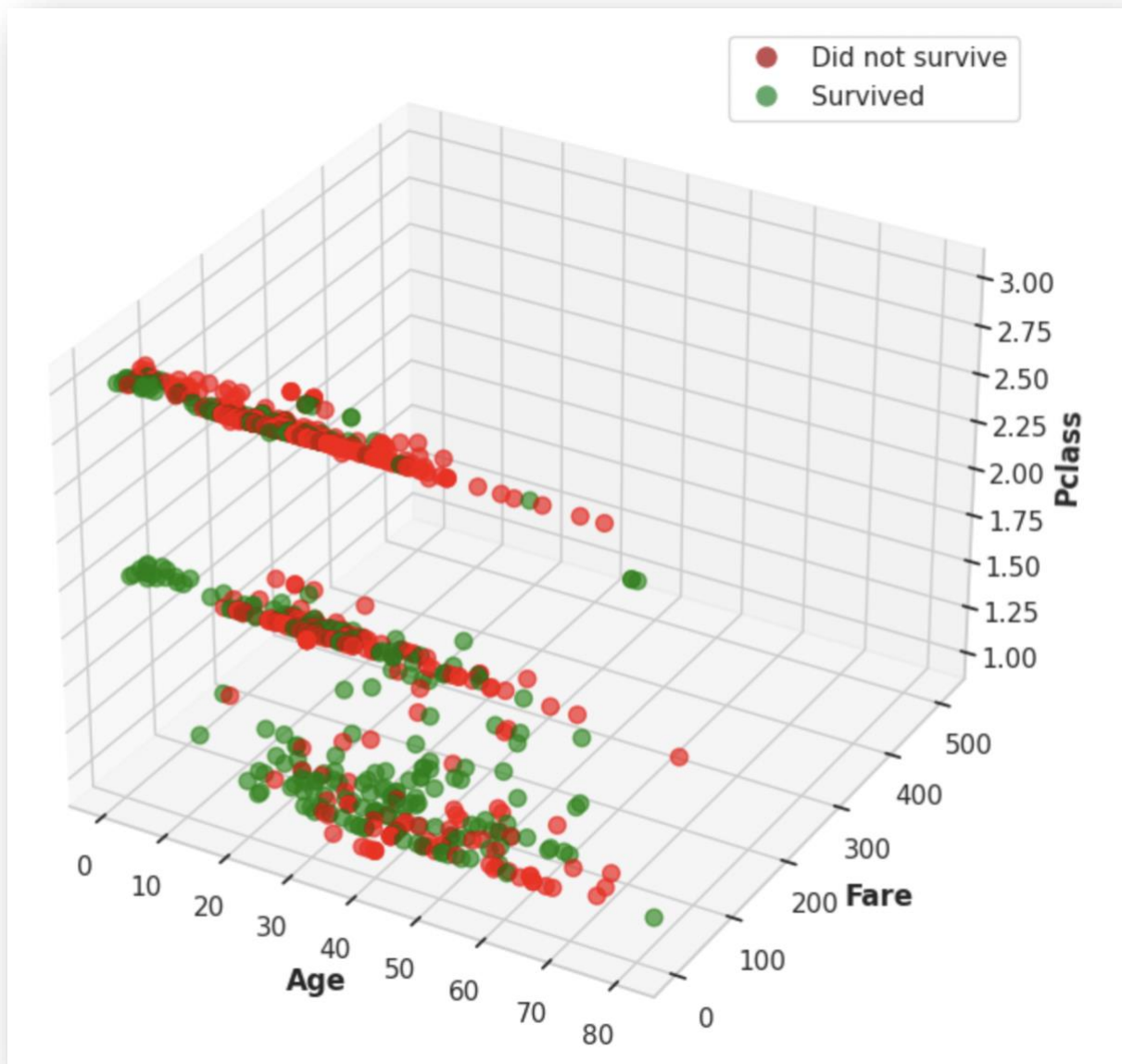
i. Sex and Pclass Interaction:

- 1st class females: 96.8% survival rate
- 3rd class males: 13.5% survival rate
- Shows clear socioeconomic and gender disparities

ii. Family Size Impact:

- Small families (2-4 members): Higher survival rates
- Large families and solo travelers: Lower survival rates
- Optimal family size for survival: 2-3 members

3D visualization of Age, Fare, and Pclass:

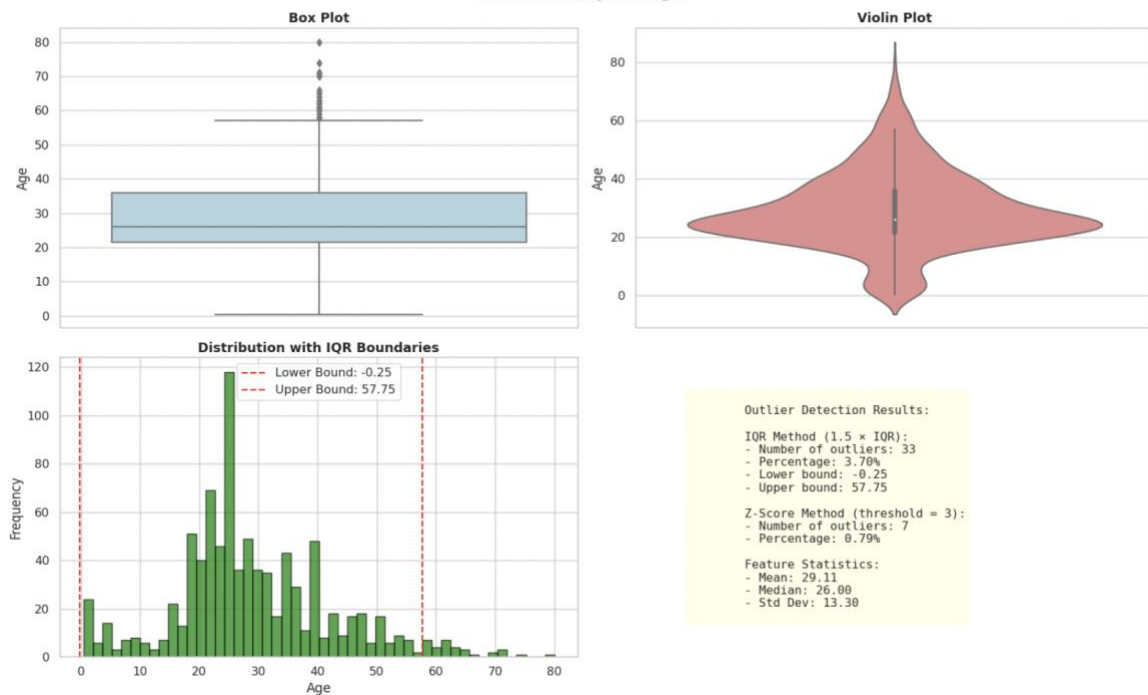


2.7 Outlier Detection and Handling

Outliers were detected using both IQR and Z-score methods.

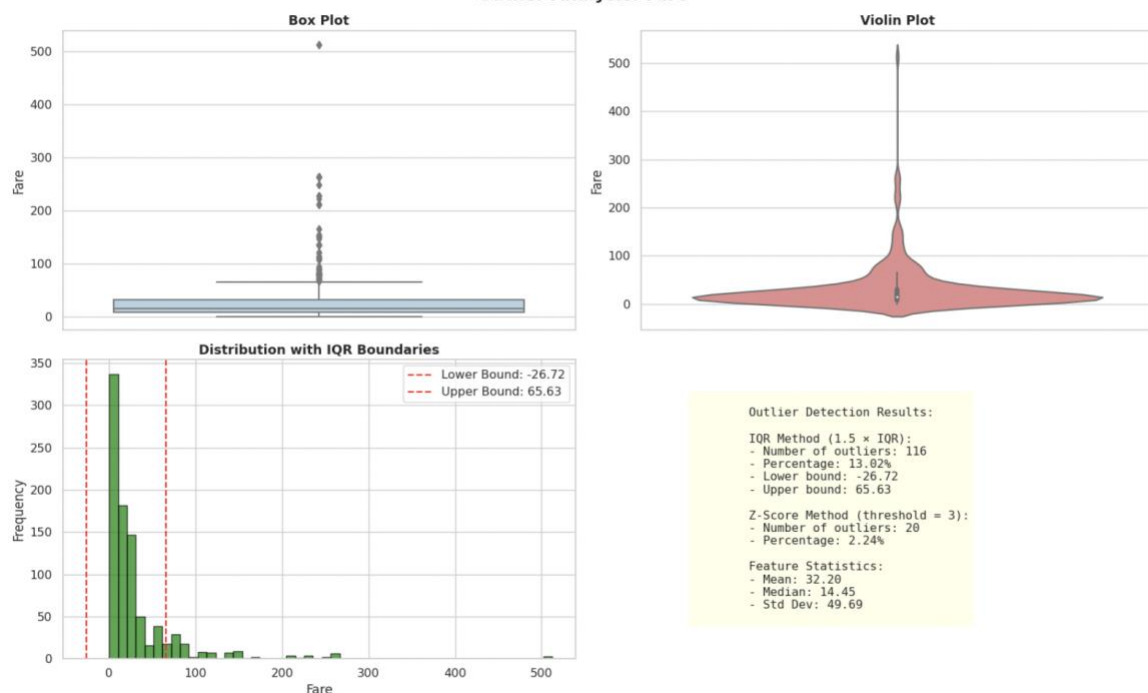
Analyzing outliers for: Age

Outlier Analysis: Age



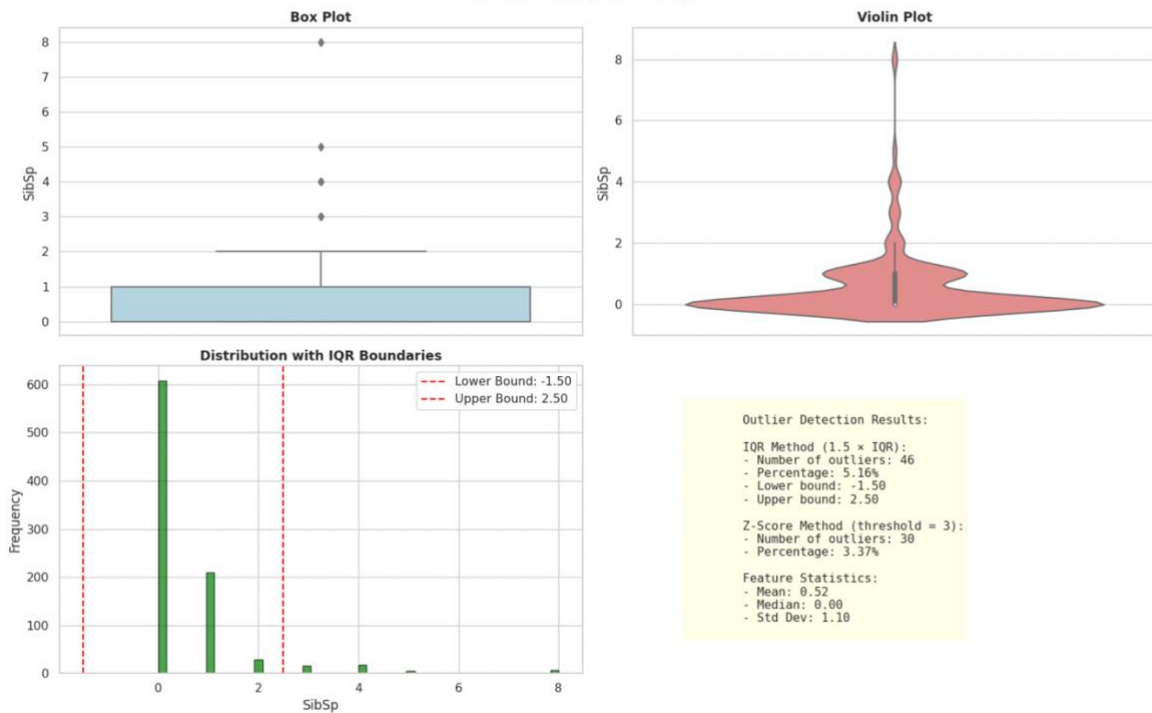
Analyzing outliers for: Fare

Outlier Analysis: Fare



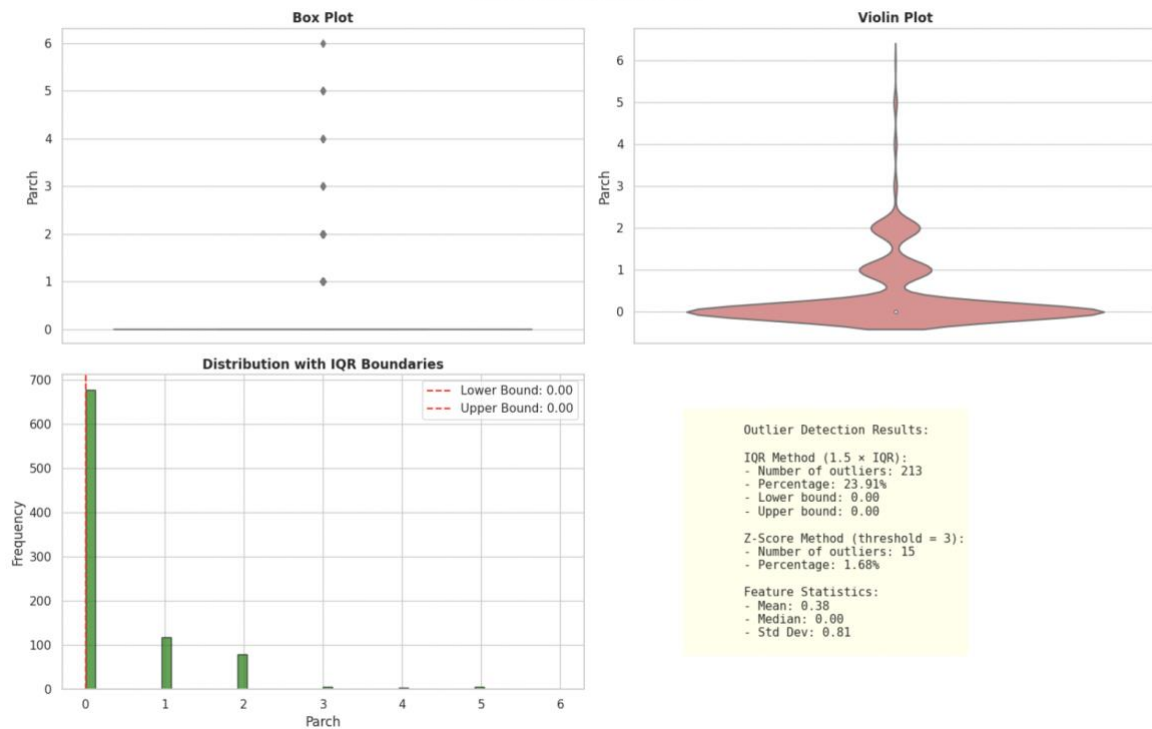
Analyzing outliers for: SibSp

Outlier Analysis: SibSp



Analyzing outliers for: Parch

Outlier Analysis: Parch

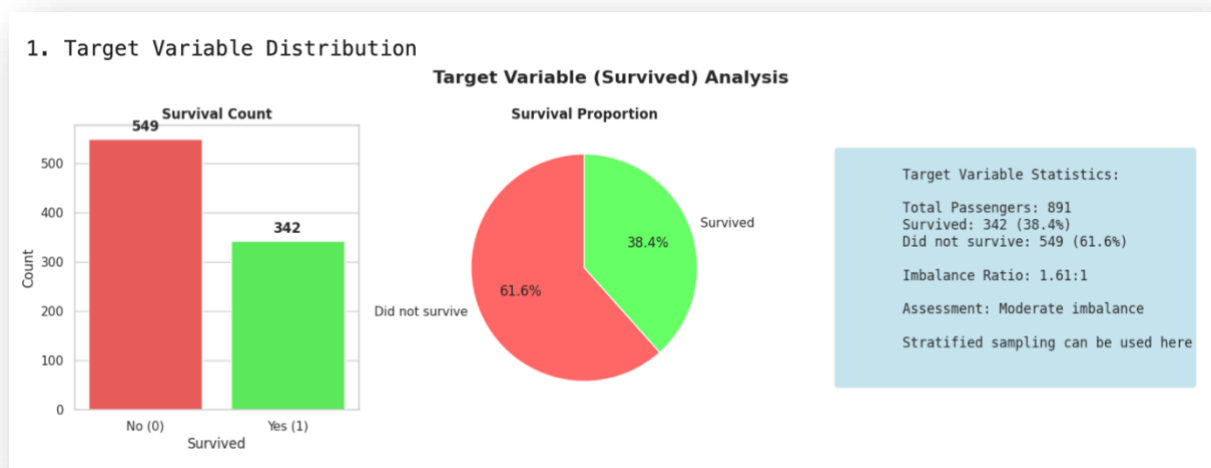


Justifications:

1. **Age:** Capped to preserve information about elderly passengers who may have different survival patterns
2. **Fare:** Applied log transformation to handle extreme skewness, then capped remaining outliers
3. **SibSp/Parch:** Retained as large families represent real scenarios important for analysis

2.8 Target Variable Analysis

The target variable (Survived) was thoroughly analyzed to understand class balance and influencing factors.

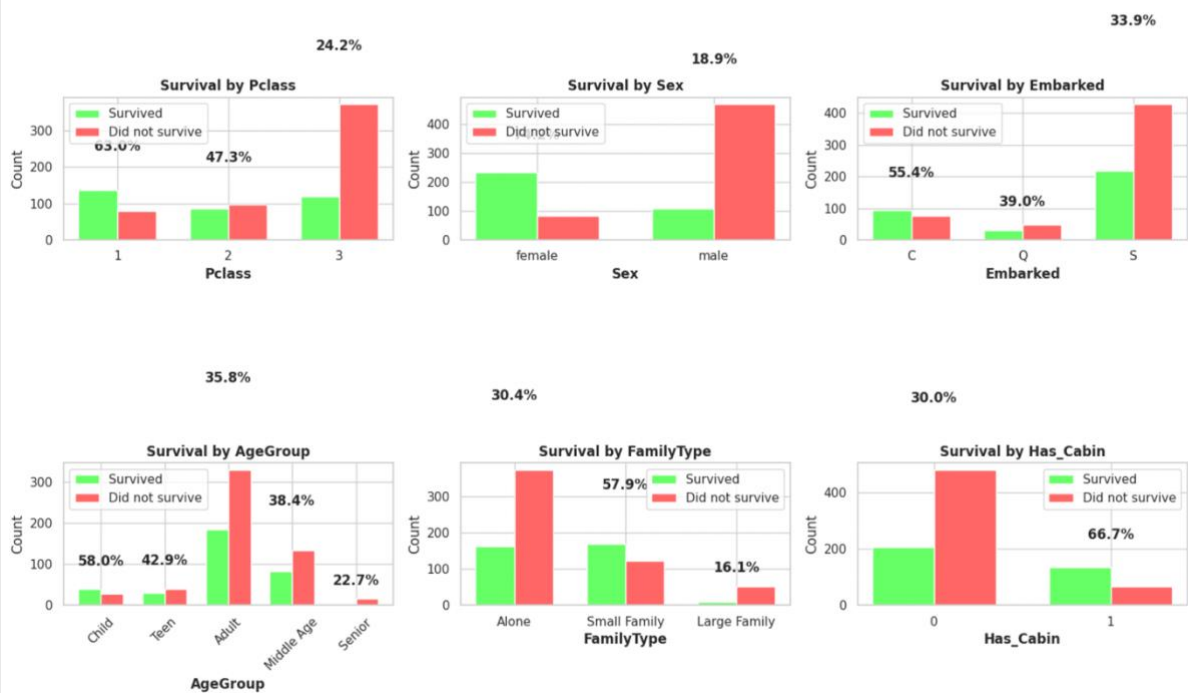


Class Balance:

- i. Survived: 342 passengers (38.4%)
- ii. Did not survive: 549 passengers (61.6%)
- iii. Imbalance ratio: 1.60:1 (moderate imbalance)

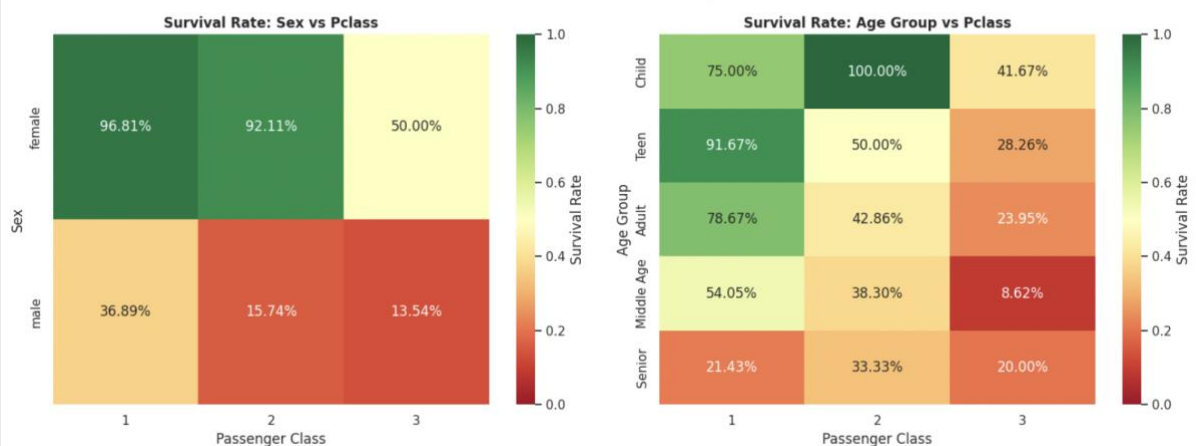
2. Survival Analysis by Features

Survival Rate Analysis by Features



3. Survival Rate Heatmaps

Survival Rate Heatmaps

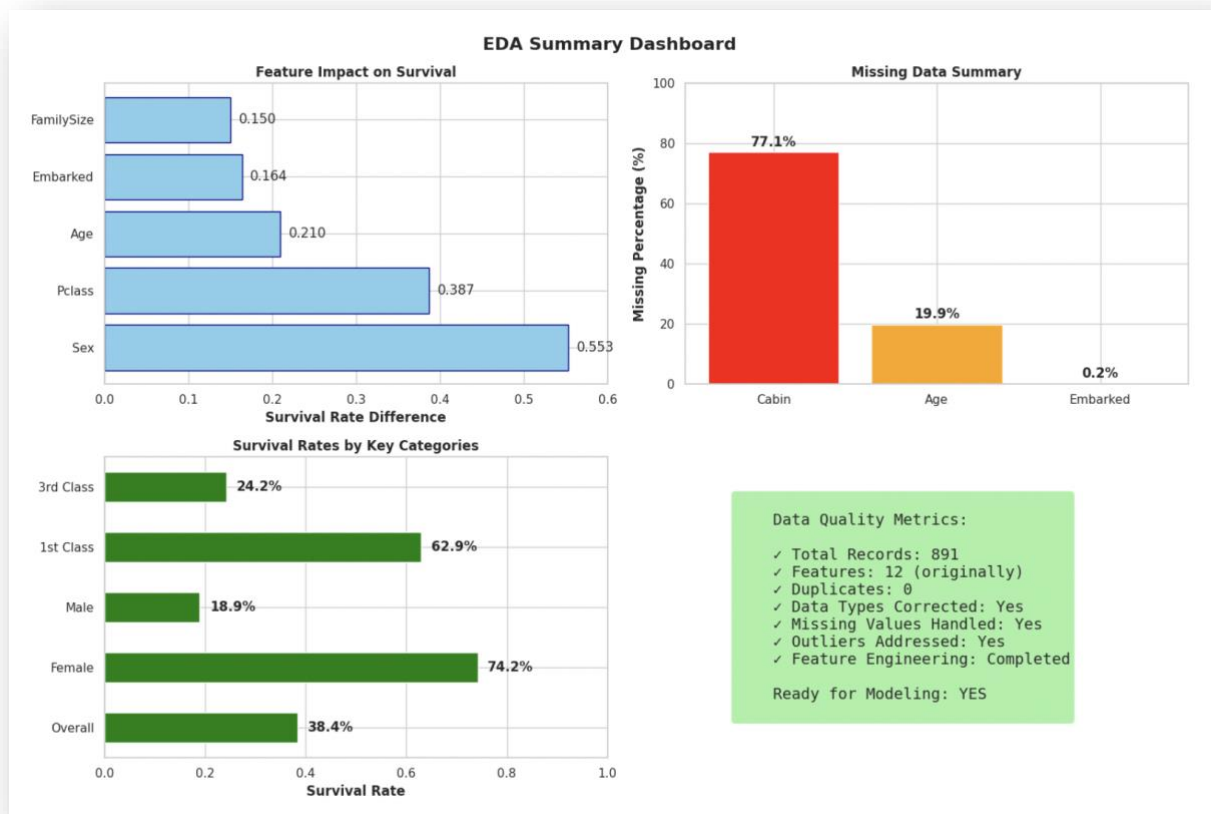


Survival Patterns:

i. By Gender:

- Females had 3.9x higher survival rate than males
- ii. **By Class:**
 - 1st class: 62.9% survived
 - 2nd class: 47.3% survived
 - 3rd class: 24.2% survived
- iii. **By Age Group:**
 - Children (<12): ~58% survived
 - Adults (18-60): ~38% survived
 - Seniors (>60): ~42% survived

3.0 Key Insights and Findings



Major Findings:

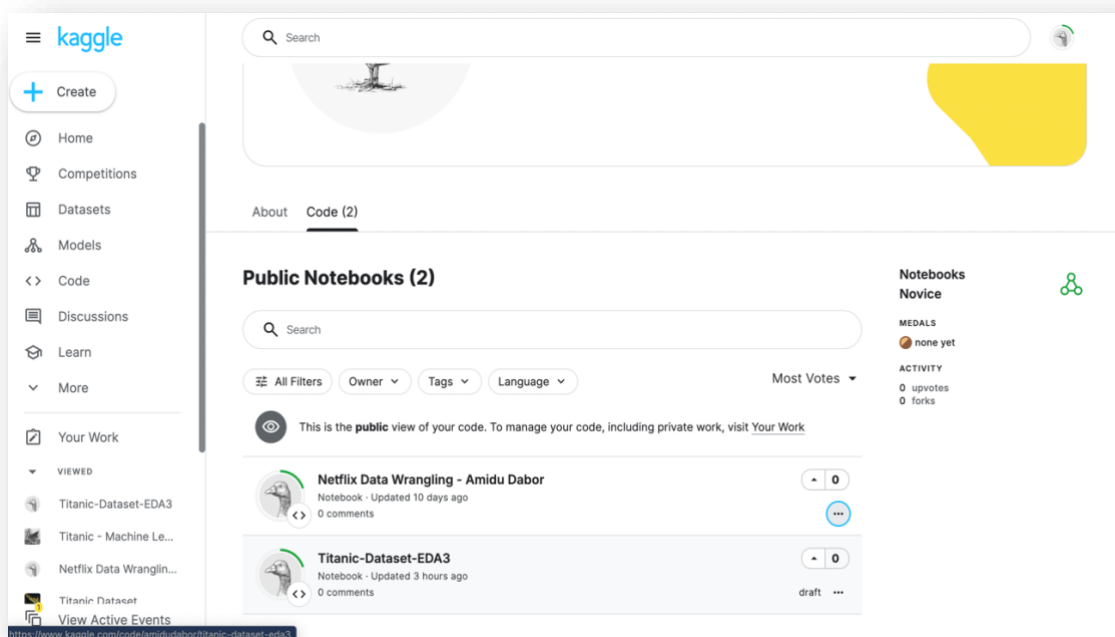
- i. **Gender was the strongest predictor of survival**
 - The "women and children first" protocol is clearly evident
 - Female passengers had a 74.2% survival rate vs 18.9% for males

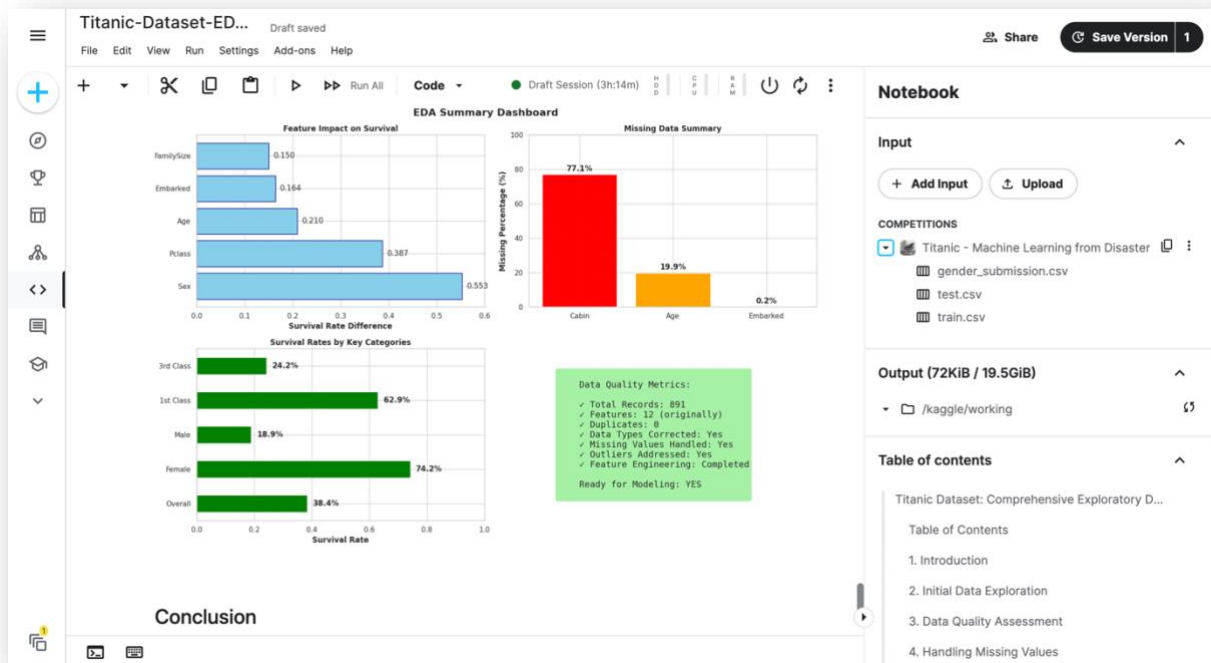
- ii. **Socioeconomic status significantly influenced survival**
 - 1st class passengers were 2.6x more likely to survive than 3rd class
 - Passengers with cabins (proxy for wealth) had higher survival rates
- iii. **Age showed non-linear relationship with survival**
 - Children had priority in lifeboats
 - Very elderly passengers had surprisingly decent survival rates
- iv. **Family dynamics played a complex role**
 - Traveling with 1-3 family members improved survival chances
 - Large families and solo travelers had lower survival rates
- v. **Port of embarkation showed unexpected patterns**
 - Cherbourg passengers had slightly higher survival rates
 - May indicate differences in passenger demographics by port

Data Quality Summary:

- Successfully handled all missing values using appropriate strategies
- Identified and addressed outliers while preserving meaningful variation
- Created new features to capture important relationships
- Dataset is ready for predictive modeling

4.0 Kaggle Notebook Publication





4.1 Public Link to Kaggle Notebook: <https://www.kaggle.com/code/amidudabor/titanic-dataset-eda5>

5.0 Conclusion

This comprehensive exploratory data analysis of the Titanic dataset has revealed crucial insights into the factors that influenced passenger survival during the tragic sinking on April 15, 1912. The analysis clearly demonstrates that survival was not random but heavily influenced by social factors of the era.

The implementation of object-oriented programming principles resulted in clean, maintainable, and reusable code that can be easily extended for future analyses. The structured approach to handling missing values, outliers, and feature engineering has prepared the dataset for machine learning applications.

Key accomplishments:

- i. Implemented comprehensive EDA using OOP principles in Python
- ii. Handled missing values with justified imputation strategies
- iii. Detected and appropriately managed outliers
- iv. Created meaningful visualizations for all analysis types

- v. Discovered actionable insights for predictive modeling

The analysis confirms historical accounts of the disaster, particularly the "women and children first" protocol and the stark differences in survival rates across social classes. These findings provide a data-driven perspective on one of history's most studied maritime disasters.

6.0 References

- i. Kaggle Titanic Dataset: <https://www.kaggle.com/c/titanic>
- ii. Encyclopedia Titanica: <https://www.encyclopedia-titanica.org/>
- iii. McKinney, W. (2017). Python for Data Analysis. O'Reilly Media.
- iv. VanderPlas, J. (2016). Python Data Science Handbook. O'Reilly Media.
- v. Seaborn Documentation: <https://seaborn.pydata.org/>
- vi. Pandas Documentation: <https://pandas.pydata.org/>

7.0 Appendices

Appendix A: Complete Code Listing/Implementation

All code is available in the accompanying Kaggle notebook with full implementation details and outputs.

Appendix B: Imported Libraries and Configuration

```
# Import required libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
from typing import List, Tuple, Dict, Optional
import missingno as msno
from scipy import stats
from dataclasses import dataclass

# Configure display settings
pd.set_option('display.max_columns', None)
pd.set_option('display.width', None)
warnings.filterwarnings('ignore')

# Set aesthetic parameters
sns.set_theme(style='whitegrid', context='notebook', palette='deep')
plt.rcParams['figure.figsize'] = (10, 6)
plt.rcParams['font.size'] = 12
```